

Dummy variables!

Iván Andrés Trujillo

ivantrujillo1229

ai-page.readthedocs.io

In this section we are going to see the dummy variables

Suppose the model

$$y_i = \beta_0 + \beta_1 D_{1i} + \epsilon_i$$

$$D_{1i} = \begin{cases} 1, & \text{if } i\text{-th observation present a feature} \\ 0, & \text{Otherwise.} \end{cases}$$

therefore

$$E(y \mid D_1 = 0) = \beta_0$$

$$E(y \mid D_1 = 1) = (\beta_0 + \beta_1)$$

Remember that $E(\epsilon_i) = 0$.

Note therefore that

$$\beta_1 = E(y \mid D = 1) - E(y \mid D = 0)$$

The model:

$$y = \beta_0 + \beta_1 D_1 + \beta_2 x + \epsilon \quad (1)$$

$$D_1 = \begin{cases} 1, & \text{If i-th present the feature A} \\ 0, & \text{Otherwise.} \end{cases}$$

And x is a continuous variable.

E(y x)	D1	
	0	1
	$\beta_0 + \beta_2 x$	$(\beta_0 + \beta_1) + \beta_2 x$

$$\frac{dE(y \mid x)}{dx} = \beta_2, \quad D_1 = 0$$
$$\frac{dE(y \mid x)}{dx} = \beta_2, \quad D_1 = 1$$

The effect of x on y doesn't depend the level of D_1 .

The model:

$$y = \beta_0 + \beta_1 D_1 + \beta_2 x + \beta_3 (x D_1) + \epsilon \quad (2)$$

$$D_1 = \begin{cases} 1, & \text{If } i\text{-th present the feature A} \\ 0, & \text{Otherwise.} \end{cases}$$

And x is a continuous variable.

E(y x)	D1	
	0	1
	$\beta_0 + \beta_2 x$	$(\beta_0 + \beta_1) + (\beta_2 + \beta_3)x$

$$\frac{dE(y | x)}{dx} = \beta_2, \quad D_1 = 0$$

$$\frac{dE(y | x)}{dx} = \beta_2 + \beta_3, \quad D_1 = 1$$

The effect of x on y depend the level of D_1 .

This is equivalent to run to different regressions in the subpopulations (*why?*)

If the feature A dont have importante then we can check if

$$\beta_1 = \beta_3 = 0$$

The model:

$$y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \epsilon \quad (3)$$

$$D_1 = \begin{cases} 1, & \text{If } i\text{-th present the feature A} \\ 0, & \text{Otherwise.} \end{cases}$$

$$D_2 = \begin{cases} 1, & \text{if } i\text{-th present the feature B} \\ 0, & \text{Otherwise.} \end{cases}$$

E(y)		D2	
		0	1
D1	0	β_0	$\beta_0 + \beta_2$
	1	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2$

Now see

$$\begin{aligned} &E(y \mid D_1 = 1, D_2 = 1) - E(y \mid D_1 = 1, D_2 = 0) \\ &= \beta_0 + \beta_1 + \beta_2 - (\beta_0 + \beta_1) \\ &= \beta_2 \end{aligned}$$

$$\begin{aligned} &E(y \mid D_1 = 0, D_2 = 1) - E(y \mid D_1 = 0, D_2 = 0) \\ &= \beta_0 + \beta_2 - \beta_0 \\ &= \beta_2 \end{aligned}$$

The effect on Y of the feature B ($D_1 = 1$) is independent of the presence of the feature A ($D_2 = 1$).

The model:

$$y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 (D_1 D_2) + \epsilon \quad (4)$$

$$D_1 = \begin{cases} 1, & \text{If } i\text{-th present the feature A} \\ 0, & \text{Otherwise.} \end{cases}$$

$$D_2 = \begin{cases} 1, & \text{if } i\text{-th present the feature B} \\ 0, & \text{Otherwise.} \end{cases}$$

		D2	
		0	1
D1	0	β_0	$\beta_0 + \beta_2$
	1	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

Now see

$$\begin{aligned} &E(y \mid D_1 = 1, D_2 = 1) - E(y \mid D_1 = 1, D_2 = 0) \\ &= \beta_0 + \beta_1 + \beta_2 + \beta_3 - (\beta_0 + \beta_1) \\ &= \beta_2 + \beta_3 \end{aligned}$$

$$\begin{aligned} &E(y \mid D_1 = 0, D_2 = 1) - E(y \mid D_1 = 0, D_2 = 0) \\ &= \beta_0 + \beta_2 - \beta_0 \\ &= \beta_2 \end{aligned}$$

β_3 is the differential effect of D_2 (feature B) according to the presence or absence of feature A .

Frame Title

The model:

$$y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 (D_1 D_2) + \beta_4 x + \beta_5 (x D_1) + \epsilon \quad (5)$$

$$D_1 = \begin{cases} 1, & \text{If } i\text{-th present the feature A} \\ 0, & \text{Otherwise.} \end{cases}$$

$$D_2 = \begin{cases} 1, & \text{if } i\text{-th present the feature B} \\ 0, & \text{Otherwise.} \end{cases}$$

E(y)		D2	
		0	1
D1	0	$\beta_0 + \beta_4 x$	$\beta_0 + \beta_2 + \beta_4 x$
	1	$\beta_0 + \beta_1 + \beta_4 x + \beta_5 x$	$\beta_0 + \beta_1 + \beta_2 + \beta_3 + \beta_4 x + \beta_5 x$

Now see

$$\begin{aligned} & E(y \mid D_1 = 1, D_2 = 1) - E(y \mid D_1 = 1, D_2 = 0) \\ &= \beta_0 + \beta_1 + \beta_2 + \beta_3 + \beta_4 x + \beta_5 x - (\beta_0 + \beta_1 + \beta_4 x + \beta_5 x) \\ &= \beta_2 + \beta_3 \end{aligned}$$

$$\begin{aligned} & E(y \mid D_1 = 0, D_2 = 1) - E(y \mid D_1 = 0, D_2 = 0) \\ &= \beta_0 + \beta_2 + \beta_4 x - (\beta_0 + \beta_4 x) \\ &= \beta_2 \end{aligned}$$

β_3 is the differential effect of D_2 (feature B) according the presence of absence of feature A .

$$\frac{dE(y | x)}{dx} = \beta_4, D_1 = 0, D_2 = 0.$$

$$\frac{dE(y | x)}{dx} = \beta_4 + \beta_5, D_1 = 1, D_2 = 0.$$

$$\frac{dE(y | x)}{dx} = \beta_4, D_1 = 0, D_2 = 1.$$

$$\frac{dE(y | x)}{dx} = \beta_4 + \beta_5, D_1 = 1, D_2 = 1.$$

Note that D_2 is not take into account.